Exploratory Data analysis insights

Summary statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.00	3951.0000	7153.000	9552.0000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.00	5793.5000	18761.500	25514.2500	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.36	6.8400	10.455	19.4825	5.681810e+03
ActualPrice	10692.0	35.643671	148.246016	0.49	10.9900	15.990	28.9900	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.00	750.0000	750.000	750.0000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.00	36.0000	262.000	1975.7500	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.71	453.4575	3655.465	20738.2450	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.00	33.0000	261.000	1929.2500	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.00	729.2200	5298.045	28396.9150	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.00	289.7100	2857.800	16059.5625	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.00	4.8000	46.570	418.6500	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.09	14069.8700	50293.620	79528.9900	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.78	52.9200	1399.640	8660.2000	1.290668e+06
ProfitMargin	10692.0	-15.620805	443.555306	-23730.64	13.3275	30.405	39.9600	9.972000e+01
StockTurnover	10692.0	1.706770	6.020480	0.00	0.8100	0.980	1.0400	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504371	8.459091	0.00	1.1500	1.440	1.6700	3.529300e+02

Max StockTurnover - 274.5
Max PurchasePrice - 5681.81
Max ActualPrice - 7499.99
Max FreightCost - 257032.07000000007

Negative & Zero Values:

Gross Profit: Minimum of -52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Had a minimum of -∞ which was rectified but it suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

Freight Cost: Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

Stock Turnover: Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- **Gross Profit ≤ 0** (to exclude transactions leading to losses).
- **Profit Margin ≤ 0** (to ensure analysis focuses on profitable transactions).
- **Total Sales Quantity = 0** (to eliminate inventory that was never sold)

Research Question & key findings

1. Brands for promotional or pricing adjustment

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Brands with Low Sales but High Profit Margins:
                      Description  TotalSalesDollars  ProfitMargin
6       12 Days of Pearls Gift Set             309.69         97.68
45                  4 Orange Vodka             483.78         85.90
57       A Bichot Clos Marechaudes             539.94         67.74
59          A Bichot Merc Champs M             515.88         94.55
96      Absolut Orient Apple Vodka             119.94         87.65
...                            ...                ...           ...
7588  Woodford Rsv Master Coll Pnt             489.95         73.76
7663   Zardetto Pros di Coneg Brut             345.86         88.76
7677         Zerran Tinto Montsant             139.91         77.59
7686         Zhenka Vodka 80 Proof             240.39         87.30
7696                     Zorah Red             417.81         67.56
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198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

2. Top Vendors by Purchase Contribution

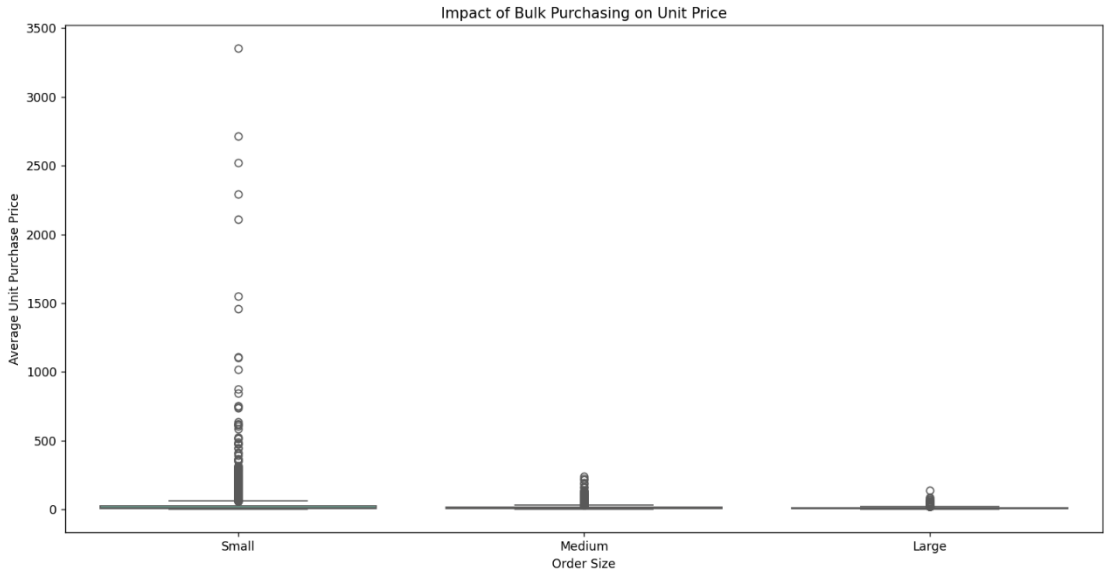
The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.

	VendorName	TotalPurchaseDollars	GrossProfit	TotalSalesDollars	PurchaseContribution%	cumulative_contribution
25	DIAGEO NORTH AMERICA INC	50097226.16	1.789287e+07	6.799010e+07	16.30	16.30
57	MARTIGNETTI COMPANIES	25502095.83	1.382826e+07	3.933036e+07	8.30	24.60
68	PERNOD RICARD USA	23851164.17	8.212032e+06	3.206320e+07	7.76	32.36
46	JIM BEAM BRANDS COMPANY	23494304.32	7.928716e+06	3.142302e+07	7.64	40.00
6	BACARDI USA INC	17432020.26	7.422797e+06	2.485482e+07	5.67	45.67
20	CONSTELLATION BRANDS INC	15273708.08	8.945038e+06	2.421875e+07	4.97	50.64
11	BROWN-FORMAN CORP	13238939.18	5.008291e+06	1.824723e+07	4.31	54.95
30	E & J GALLO WINERY	12068539.22	6.331360e+06	1.839990e+07	3.93	58.88
106	ULTRA BEVERAGE COMPANY LLP	11167081.61	5.335463e+06	1.650254e+07	3.63	62.51
53	M S WALKER INC	9764312.60	4.942146e+06	1.470646e+07	3.18	65.69

3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost ($10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

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OrderSize
Small     39.068186
Medium    15.486414
Large     10.777625
Name: UnitPurchasePrice, dtype: float64
```

4. Identifying Vendors with highest unsold inventory

Total Unsold Inventory Capital: $2.71M

High unsold inventory ($2.71M) is concentrated among a few vendors such as Diageo North America Inc and Jim Beam Brands Company, indicating overstocking or demand mismatch. This ties up capital and increases holding

costs. Targeted inventory optimization and demand-driven purchasing can significantly improve efficiency.

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722209.05
46	JIM BEAM BRANDS COMPANY	554665.63
68	PERNOD RICARD USA	470625.61
116	WILLIAM GRANT & SONS INC	401960.83
30	E & J GALLO WINERY	228282.61
79	SAZERAC CO INC	198436.41
11	BROWN-FORMAN CORP	177733.74
20	CONSTELLATION BRANDS INC	133617.62
61	MOET HENNESSY USA INC	126477.70
77	REMY COINTREAU USA INC	118598.15

5. Identifying vendors with low inventory turnover

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

VendorName	
TAMWORTH DISTILLING	0.800000
WALPOLE MTN VIEW WINERY	0.820000
SWEETWATER FARM	0.853333
BRONCO WINE COMPANY	0.867500
SWEET BABY VINEYARD	0.875000
AMERICAN SPIRITS EXCHANGE	0.880000
FULCHINO VINEYARD INC	0.880000
SOUTHERN GLAZERS W&S OF NE	0.883333
CENTEUR IMPORTS LLC	0.886250
CALEDONIA SPIRITS INC	0.897500

Final Recommendation

- Reassess pricing strategies for low-sales, high-margin brands to improve sales volume while maintaining healthy profit margins.
- Diversify vendor partnerships to reduce dependency on a limited number of suppliers and mitigate supply chain risks.
- Leverage bulk purchasing strategies to maintain cost advantages while ensuring efficient inventory management.
- Optimize slow-moving inventory through demand-aligned purchasing, targeted clearance strategies, and improved stock planning.
- Strengthen marketing and distribution efforts for low-performing vendors to increase sales volume without compromising profit margins.
- Implementing these strategies can enhance operational efficiency, improve cash flow, reduce risks, and drive sustainable profitability.